

Tata Steel Hackathon: Predicted Final Interview Flow

Phase 1: The Introduction (2-3 Minutes)

The judges will want a high-level overview before diving into the code.

Q: "Vikhil, congratulations on making it to the finals. To start, can you give us a 2-minute overview of your entire project, covering Round 1 and Round 2?"

How to Answer: "I built a two-part Industrial AI solution. For Round 1, I built a highly optimized Random Forest classifier to detect the Alpha Defect from tabular sensor data. For Round 2, I built a fully local GenAI Assistant. It features a Hugging Face Vision model to diagnose broken equipment from photos, a local RAG pipeline to search safety manuals, and a conversational agent that automatically generates inventory requests."

Phase 2: Round 1 Deep Dive - Data Science (5-7 Minutes)

The Data Science experts will test your math and logic from Round 1.

Q: "We required 100% Recall for the Alpha Defect. How exactly did you achieve that?"

How to Answer: "I optimized the probability classification threshold of my Random Forest model, making it hyper-sensitive to the defect to ensure zero false negatives."

Q: "Why did you choose Pseudo-Labeling for your data augmentation?"

How to Answer: "The Alpha defect is extremely rare. By using semi-supervised learning (Pseudo-Labeling), I merged highly confident predictions with my training set to give the model more examples of the underlying defect distribution."

Phase 3: Round 2 Deep Dive - GenAI & Architecture (5-7 Minutes)

The Software Architects will test your engineering from Round 2.

Q: "Why did you choose local models like Ollama instead of using the OpenAI API?"

How to Answer: "Data Privacy. Sending proprietary plant schematics or equipment issues to OpenAI is a massive security risk. Running Ollama locally ensures 100% data security."

Q: "How does your automated inventory workflow handle Supervisor approval?"

How to Answer: "The AI acts as an assistant, not a manager. It generates a request trigger that hits the SQLite database. But before any parts are deducted, a notification is sent to the Supervisor portal so a human can physically verify the issue and approve the budget."

Phase 4: The 'Bridging' Question (The Ultimate Test)

They will ask a question to see if you can connect both rounds together.

Q: "In Round 1 you built a tabular model to detect defects. In Round 2, you built an AI chatbot with Vision.

Tata Steel Hackathon: Predicted Final Interview Flow

How would you combine these two systems in a real Tata Steel plant?"

How to Answer: "The Round 1 Random Forest model would run silently in the background, analyzing the X1-X49 sensor data in real-time. If it detects an Alpha Defect, it would trigger an automated alert to the Round 2 AI Chatbot. The Chatbot would notify the Engineer on duty, pull the exact safety SOP from ChromaDB, and ask the Engineer to upload a photo of the machine for the Vision model to double-check the damage. This creates a complete end-to-end autonomous monitoring and diagnostic system."